

Failure Theories:

①

Introduction:

When an external load is applied, stresses and strains are produced in the body. These follow Hooke's law up to the elastic limit.

However, if the stresses are greater and if they exceed the elastic limit, permanent deformations occur in the body. Whenever permanent deformations occur the body is said to have 'failed'. This also indicates that failure is not the rupture/breakage of the body/specimen.

Need for failure theories:

1. To design structural components and calculate margin of safety.
2. Guide material developments.
3. Determine weak & strong directions.

Now, let us consider the failure of bar in a tensile test. Stress is directly proportional to strain up to the elastic limit. Beyond the elastic limit if the stress (tensile stress increases) the bar will fail. Why does this failure occur? This could be due to shear stresses, strain energy etc. Many theories have been developed to explain the cause of failure in materials.

Static failure Theories.

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- ① Maximum principal stress Theory.
- ② Maximum principal strain Theory.
- ③ Maximum shear stress theory.
- ④ Maximum strain Energy Theory.
- ⑤ Maximum shear strain energy theory.

Note:

σ_1 , σ_2 and σ_3
will be used as
principal stresses in
a complex system.

Maximum principal Stress Theory:-

Also called as 'Rankine's Theory'. This theory states that a material in a complex state of stress fails, when the maximum principal stress in the material reaches the value of stress at an elastic limit.

This theory was developed in the year 1850.

Crack will begin to initiate at the most highly stressed point in a brittle material when the largest principal stress at that point reaches ultimate stress.

This criteria/Theory has good experimental verification even though it assumes ultimate strength is same in compression and tension.

* This theory has very poor agreement with experimentation for yielding i.e. for ductile materials.

But has been very successfully used for brittle materials.

Limitations:-

- ① Does not distinguish between tension and compression.
- ② Does not depend on the orientation of the principal plane.

If σ_1, σ_2 and σ_3 are principal stresses in a point in 3 mutually perpendicular directions and σ_1 and σ_2 are tensile σ_3 is compressive and $\sigma_1 > \sigma_2$. If σ_t and σ_c are tensile stress at ultimate for brittle materials in tension and compression

Then according to this theory, the component will fail if $\sigma_1 \geq \sigma_t$ and $\sigma_3 \geq \sigma_c$ in tension and compression respectively.

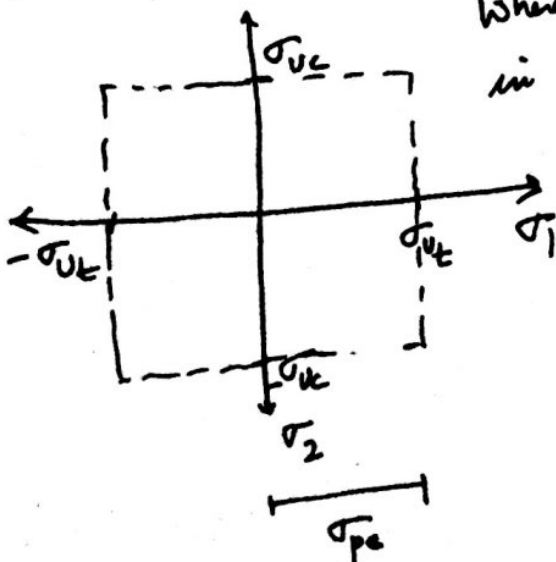
In a 2-D state of stress, principal stress is given by.

$$\sigma_1, \sigma_2 = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

Graphical representation of the theory in 2-Dimensions:-

Figure shows failure Envelope:-

Where σ_{uc} and σ_{ut} are Ultimate stress in compression and tension.



Note: σ_1 & σ_2 in first & 3rd Quadrant are +ve and -ve.

And hence, $\sigma_1 = \pm \sigma_u$ and $\sigma_2 = \pm \sigma_u$

Maximum principal strain theory

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Also called ~~Coulomb's Theory~~ St. Venant's Theory.

According to this theory failure occurs when the maximum principal strain reaches the strain value in Uniaxial stress at elastic limit.

When the minimum principal strain (Compressive strain max) reaches strain at yield stress in compression the material will fail. Assuming ϵ_1 and ϵ_2 to be tensile strains.

$$\epsilon_1 = \frac{\sigma_1}{E} - \mu \frac{\sigma_2}{E} - \frac{\mu \sigma_3}{E}, \quad \epsilon_1 = \frac{1}{E} [\sigma_1 - \mu(\sigma_2 + \sigma_3)]$$
$$\epsilon_3 = \frac{1}{E} [\sigma_3 - \mu(\sigma_1 + \sigma_2)]$$

Strain due to yield stress in Simple tension.

$$\epsilon_t = \frac{\sigma_t}{E}$$

$\epsilon_c = \frac{\sigma_c}{E}$ strain due to yield stress in compression.

According to this theory $\epsilon_1 \geq \frac{\sigma_t}{E}$

$$|\epsilon_3| \geq \frac{\sigma_c}{E}$$

Hence substituting $\frac{1}{E} [\sigma_1 - \mu(\sigma_2 + \sigma_3)] \geq \frac{\sigma_t}{E}$, $\sigma_1 - \mu(\sigma_2 + \sigma_3) \geq \sigma_t$

Similarly $\sigma_3 - \mu(\sigma_1 + \sigma_2) \geq \sigma_c$

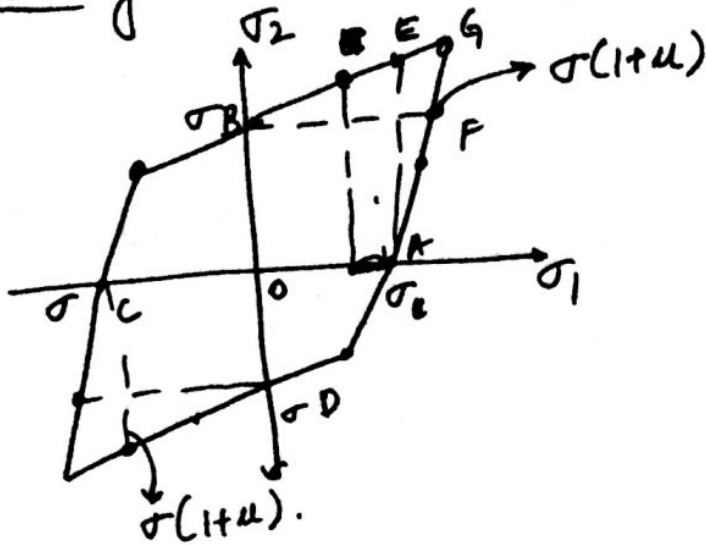
Bringing into picture the factor of safety concept-

$\frac{\sigma_t}{FOS} = \sigma_{te}$ & $\frac{\sigma_c}{FOS} = \sigma_{ce}$ where σ_{te} and σ_{ce} are working stresses (allowable) (yield) & in tension & compression.

Failure Envelope based on max principal strain Theory

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Theory



This plotting is like the equations $x = \sigma, y = \sigma$
 $y = -\sigma, x = -\sigma$

The plot of $x = \sigma$ will be a vertical line OA.
 and $x = -\sigma$ is OBC.

Maximum Shear strain Energy Theory:-

This also is called Distortion Theory.

According to this theory. The failure of the component occurs when the total shear strain energy per Unit-Volume at the elastic limit.

The total shear strain energy per Unit-Volume due to principal stresses σ_1, σ_2 and σ_3 in a stressed material is

$$= \frac{1}{12C} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]$$

In an Uniaxial stress system, the principal stresses are $\sigma_1, 0, 0$. At the elastic limit, σ_t is the tensile stress.

$\sigma_1 = \sigma_t$, hence, shear strain energy per Unit-Volume is

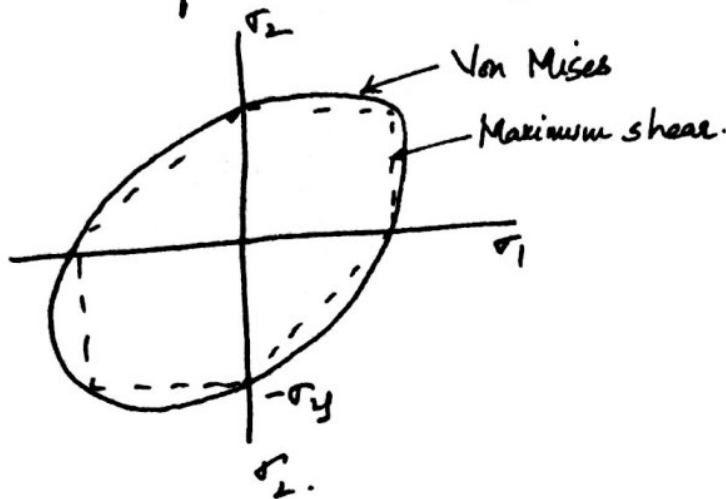
$$= \frac{1}{12C} ((\sigma_t)^2 + 0 + (0 - \sigma_t)^2) = \frac{1}{12C} [2\sigma_t^2]$$

Hence for the material to fail.

$$\frac{1}{12c} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2] \geq \frac{1}{12} [2\sigma_t^2].$$

$$[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2] \geq 2\sigma_t^2.$$

The graphical representation of the failure theory is as follows.



Maximum Shear Stress Theory:-

Also known as Coulomb's theory, or Guest's theory. According to this theory, the failure of the material will occur when the maximum shear stress in the material reaches the value of maximum shear stress in simple tension at the elastic limit.

The maximum shear stress in the material is equal to half the difference between maximum principle stress. If σ_1 , σ_2 and σ_3 are principal stresses at a point in the material for which σ_t is the principal stress in simple tension at elastic limit. then

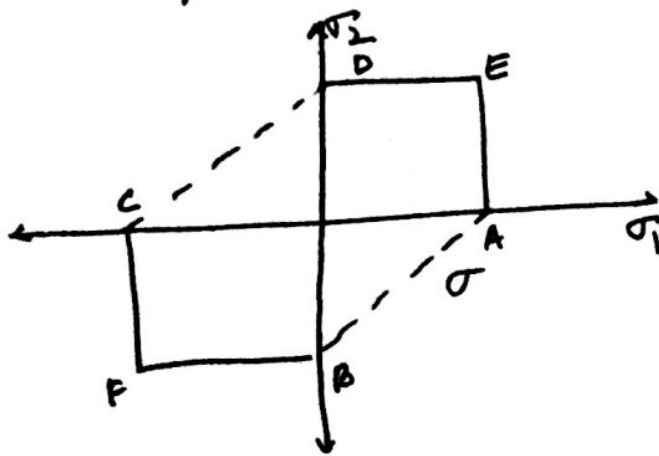
$$\text{max shear stress} = \frac{1}{2} [\sigma_1 - \sigma_3]$$

$$\therefore \text{Max Shear Stress} = \frac{1}{2} [\sigma_1 - \sigma_3] = \frac{1}{2} [\sigma_t]$$

for the failure of the material

$$\frac{1}{2} [\sigma_1 - \sigma_3] \geq \frac{1}{2} \sigma_t \quad \text{or} \quad [\sigma_1 - \sigma_3] \geq \sigma_t.$$

Graphical Representation of Max Shear Stress Theory:-



1st Quadrant

$$\sigma_1 = \sigma_t \text{ (AE)}$$

$$\sigma_2 = \sigma \text{ (DE)}$$

3rd Quadrant

$$\sigma_1 = -\sigma \text{ (CF)}$$

$$\sigma_2 = -\sigma \text{ (BF)}$$

Maximum Strain Energy Theory:-

Also known as Haigh's Theory. According to this theory, the failure of a material occurs when the total strain energy per unit volume in the material reaches strain energy per unit volume in the material at the elastic limit in simple tension.

Strain Energy in a body is equal to Work done by the load. $\frac{1}{2} P \times \delta L$

Hence Strain Energy in straining the material
 $U = \frac{1}{2} P \times \delta L.$

$$U = \frac{1}{2} \times P \times \delta L, U = \frac{1}{2} \times \sigma \times A \times \delta L \quad \left| \quad \sigma = P/A \right.$$

$$U = \frac{1}{2} \times \sigma \times A \times E \times L$$

$$\& \quad E = \frac{\delta L}{L}$$

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$$U = \frac{1}{2} \times \sigma \times E \times \underbrace{A \times L}_{\text{Volume}}, \quad U = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{Volume}.$$

$\therefore$ Strain Energy per Unit Volume

$$\frac{U}{V} = \frac{1}{2} \times \text{stress} \times \text{strain}.$$

For a 3-D system $\sigma_1, \sigma_2, \sigma_3$ are principal stresses and the corresponding strains are ϵ_1, ϵ_2 & ϵ_3 (principal strains)

$$\epsilon_1 = \frac{\sigma_1}{E} - \frac{\mu}{E} (\sigma_2 + \sigma_3), \quad \epsilon_2 = \frac{\sigma_2}{E} - \frac{\mu}{E} (\sigma_3 + \sigma_1)$$

$$\epsilon_3 = \frac{\sigma_3}{E} - \frac{\mu}{E} (\sigma_2 + \sigma_1)$$

Total strain energy per Unit-Volume in 3-D is,

$$\begin{aligned} U &= \frac{1}{2} \times \sigma_1 \times \epsilon_1 + \frac{1}{2} \times \sigma_2 \times \epsilon_2 + \frac{1}{2} \times \sigma_3 \times \epsilon_3 \\ &= \frac{1}{2} \times \sigma_1 \times \left[\frac{\sigma_1}{E} - \frac{\mu}{E} (\sigma_2 + \sigma_3) \right] + \frac{1}{2} \sigma_2 \left[\frac{\sigma_2}{E} - \frac{\mu}{E} (\sigma_3 + \sigma_1) \right] \\ &\quad + \frac{1}{2} \sigma_3 \left[\frac{\sigma_3}{E} - \frac{\mu}{E} (\sigma_2 + \sigma_1) \right] \\ &= \frac{1}{2E} \left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu (\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) \right] \end{aligned}$$

The strain Energy corresponding to stress at elastic limit is $\frac{1}{2} \sigma_E \times \epsilon_E$, $\frac{1}{2} \sigma_E \times \sigma_E / E = \frac{1}{2E} \sigma_E^2 //$

For the failure of the material,

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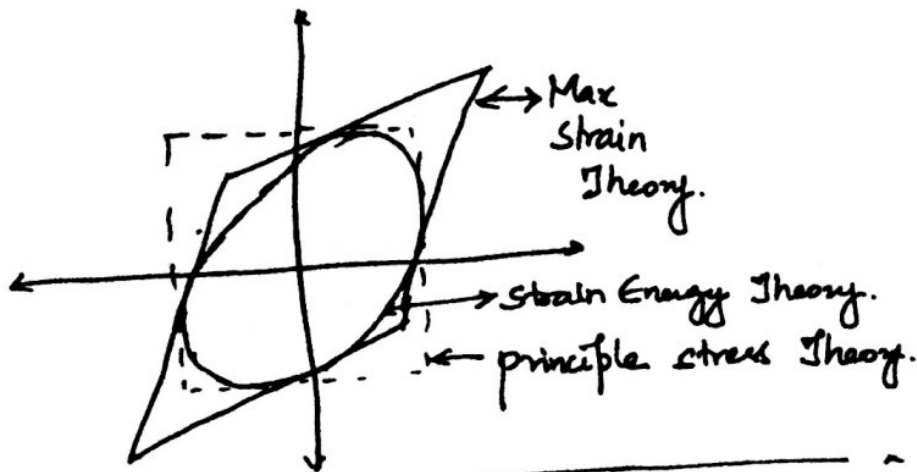
$$\frac{1}{2E} [\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3)] \geq \frac{1}{2E} \sigma_t^2$$

$$\therefore \sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3) \geq \sigma_t^2$$

For a two dimensional system.

$$\sigma_1^2 + \sigma_2^2 - 2\mu(\sigma_1\sigma_2) \geq \sigma_t^2$$

Graphical Representation of Max Strain Energy theory:-



Example problem on:- Max Principal stress & Principal strain.
Theory:-

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- ① Principal stresses at a point in an elastic material are 200 N/mm^2 (tensile), 100 N/mm^2 (tensile) and 50 N/mm^2 (compressive). If stress at elastic limit is 200 N/mm^2 . Determine if failure occurs by max principal strain Theory:- ($\mu = 0.3$)

Sol:- $\sigma_1 = 200 \text{ N/mm}^2$, $\sigma_2 = 100 \text{ N/mm}^2$, $\sigma_3 = -50 \text{ N/mm}^2$

$\sigma_L = 200 \text{ N/mm}^2$.

Failure Criteria, $= \left\{ \begin{array}{l} \sigma_1 - \mu(\sigma_2 + \sigma_3) \geq \sigma_L \\ \sigma_3 - \mu(\sigma_1 + \sigma_2) \geq \sigma_L \end{array} \right\}$.

Substituting; $200 - \mu(100 - 50) = 200 - 0.3(100 - 50)$, ~~200~~

$185 \leq 200$.

As, we can see, the criteria for failure by Max principal strain theory is not ~~not~~ satisfied, Hence, Component does not fail.

- ②. Principal stresses at a point are 100 N/mm^2 (t), 80 N/mm^2 (t) and 50 N/mm^2 (c). If stress at elastic limit is 200 N/mm^2 . Determine if failure occurs by max principal stress theory. If not determine FOS.

Sol:- Failure Criteria = $\sigma_1 \geq \sigma_L$, $\sigma_3 \geq \sigma_C$

The material does not fail as the principal stress is less than σ_L .

$\therefore$ FOS is $\sigma_p = \frac{\sigma_L}{\text{FOS}}$, $\text{FOS} = \frac{\sigma_L}{\sigma_p}$, $\frac{200}{100} = 2.0 = \text{FOS}$

Problem on Max Shear stress Theory:-

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At a section of Mild steel shaft, max torque is 1837.5 Nm and Bending Moment is 5062.5 N-m. Dia of shaft is 90 mm and stress at elastic limit is 220 N/mm².

Determine if failure will occur according to max shear stress Theory.

sol failure Criteria = $[\sigma_1 - \sigma_2] \geq \sigma_t$.

Shear stress due to Torsion

$$\tau = \frac{16T}{\pi d^3} = \frac{16 \times 1837.5}{\pi \times (0.09)^3} = 58.94 \times 10^6 \text{ N/m}^2.$$

Stress due to bending.

$$\sigma_b = \frac{My}{I}, \quad y = d/2$$

$$\sigma_b = \frac{5062.5 \times 0.045 \times 64}{\pi \times (0.09)^4} = 70.735 \times 10^6 \text{ N/m}^2.$$

Now, the principal stresses are given by.

$$\sigma_{1,2} = \frac{\sigma_b}{2} \pm \sqrt{\left(\frac{\sigma_b}{2}\right)^2 + \tau^2}$$

$$= \frac{70.735 \times 10^6}{2} \pm \sqrt{\left(\frac{70.735 \times 10^6}{2}\right)^2 + (58.94 \times 10^6)^2}$$

$$\sigma_1 = 104.11 \times 10^6 \text{ N/m}^2$$

$$\sigma_2 = -33.38 \text{ N/m}^2 \times 10^6.$$

Substituting in Criteria for failure:-

$$(104.11 \times 10^6 - (-33.38 \times 10^6)) \leq 220.$$

$\Rightarrow 137.49 \times 10^6 \geq 220$, hence material will not fail.

Problem on strain Energy theory:-

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For the given data determine if failure occurs by maximum strain Energy theory:- & Max Shear Strain Energy theory.

$$\sigma_1 = 200 \text{ N/mm}^2 \text{ (t)}$$

$$\sigma_2 = 100 \text{ N/mm}^2 \text{ (t)} \quad \mu = 0.3.$$

$$\sigma_3 = 50 \text{ N/mm}^2 \text{ (c)} \quad \sigma_t = 200 \text{ N/mm}^2.$$

Sol: ① Failure Criteria =.

$$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3) \geq \sigma_t^2.$$

Substituting,

$$200^2 + 100^2 + (-50)^2 - 2(0.3)(200 \times 100 + (100 \times -50) + 200(-50)) \\ = 49500$$

$$49500 \geq \sigma_t^2, = (200)^2$$

$$49500 \geq 40000, \text{ Material fails,}$$

The failure Criteria has been met.

$$\textcircled{2} \text{ Failure Criteria} = (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \geq 2\sigma_t^2. \\ \text{(Max Shear Strain Energy theory)}$$

$$(200 - 100)^2 + (100 - (-50))^2 + (-50 - 200)^2 = 95000$$

$$2\sigma_t^2 = 80000, \quad 95000 \geq 80000 \text{ (satisfied).}$$

The material fails as per Max Shear Strain Energy theory.